
MODEL SELECTION AND VALIDATION REPORT (remove_6_biomass model)

OBSERVATIONS:1692

SANITY: OK - all models in the selection process passed sanity checks.

SELECTED RANDOM TERM: spatiotemporal_random_fields_ar1

MODEL SUMMARY (remove_6_biomass model)

Spatiotemporal model fit by REML ['sdmTMB']

Formula: fixed_formula

Time column: year

Family: lognormal(link = 'log')

Conditional model:

	coef.est	coef.se
(Intercept)	5.51	0.15
remove_6_tric	0.56	0.03
remove_6_teve	-0.43	0.03
sbt_mean	0.26	0.09
sbt_sd	0.05	0.05
sst_sd	0.04	0.04
log_chla_mean	0.05	0.04
depth	0.21	0.05
log_sum_fishing	0.07	0.03

Dispersion parameter: 0.97

Spatiotemporal AR1 correlation (rho): 0.90

Matérn range: 185.42

Spatiotemporal marginal AR1 SD: 0.82

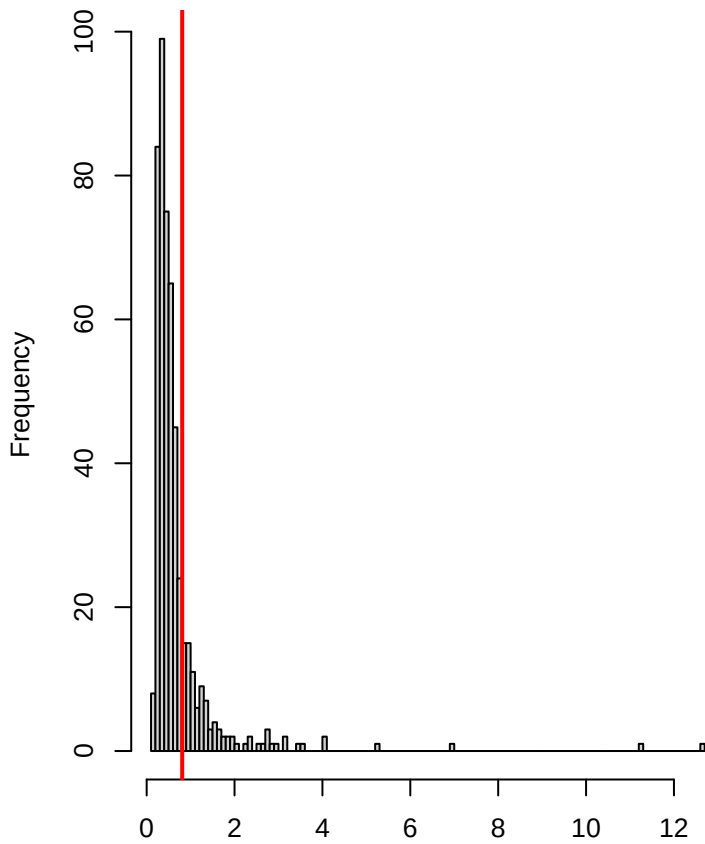
REML criterion at convergence: 10900.451

See ?tidy.sdmTMB to extract these values as a data frame.

SELECTION STATISTICS

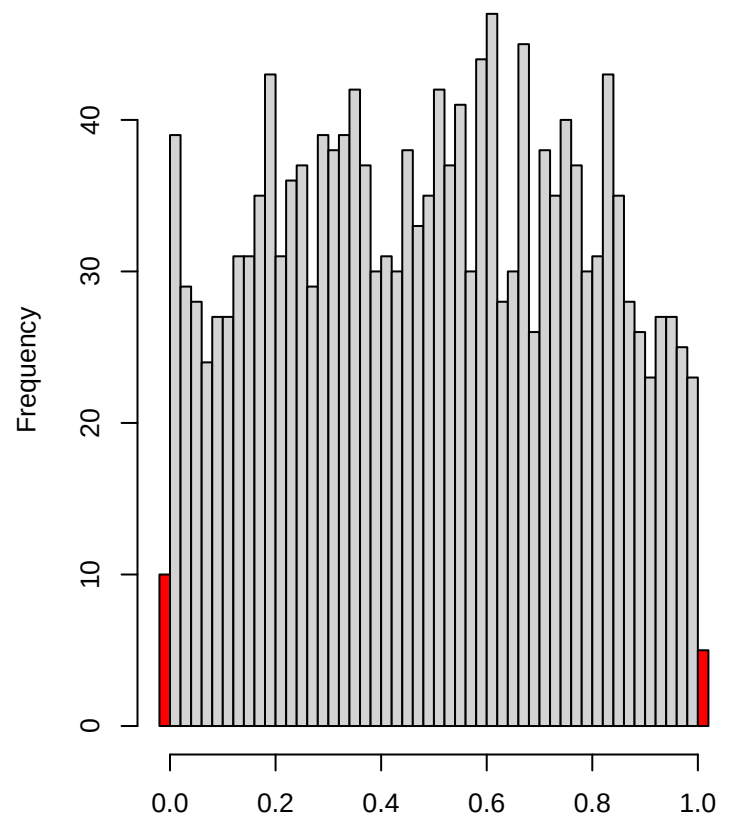
	df	AIC	delta_AIC
spatiotemporal_random_fields_ar1	13	21826.90	1.893284
spatiotemporal_random_fields_ar1_with_anisotropy	15	21828.80	71.658721
spatial_random_fields	12	21900.45	3.880694
spatial_random_fields_with_anisotropy	14	21904.34	77.958741
spatiotemporal_random_fields_iid_with_anisotropy	14	21982.29	5.199435
spatiotemporal_random_fields_iid	12	21987.49	110.956658
error_only	10	22098.45	NA

DHARMa nonparametric dispersion test via sd of residuals fitted vs. simulated



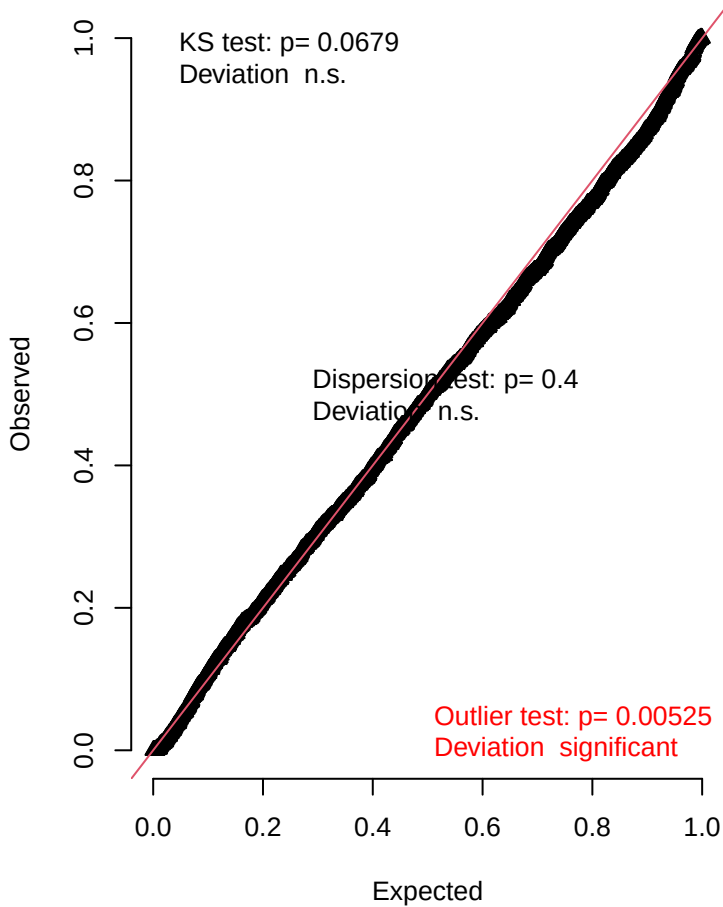
Simulated values, red line = fitted model. p-value (two.sided)

Outlier test significant

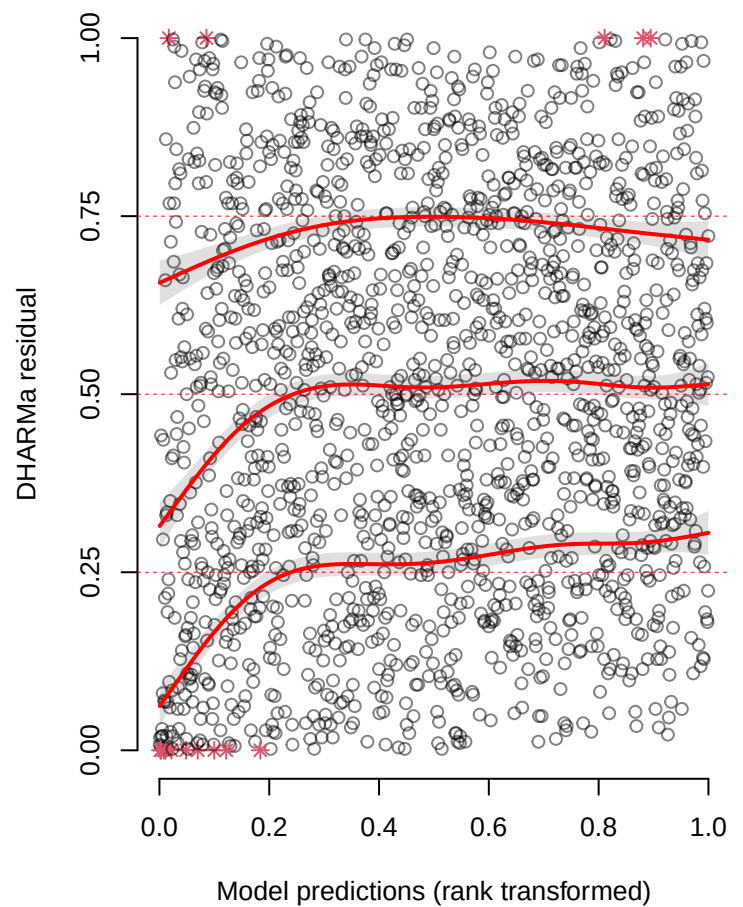


Residuals (outliers are marked red)

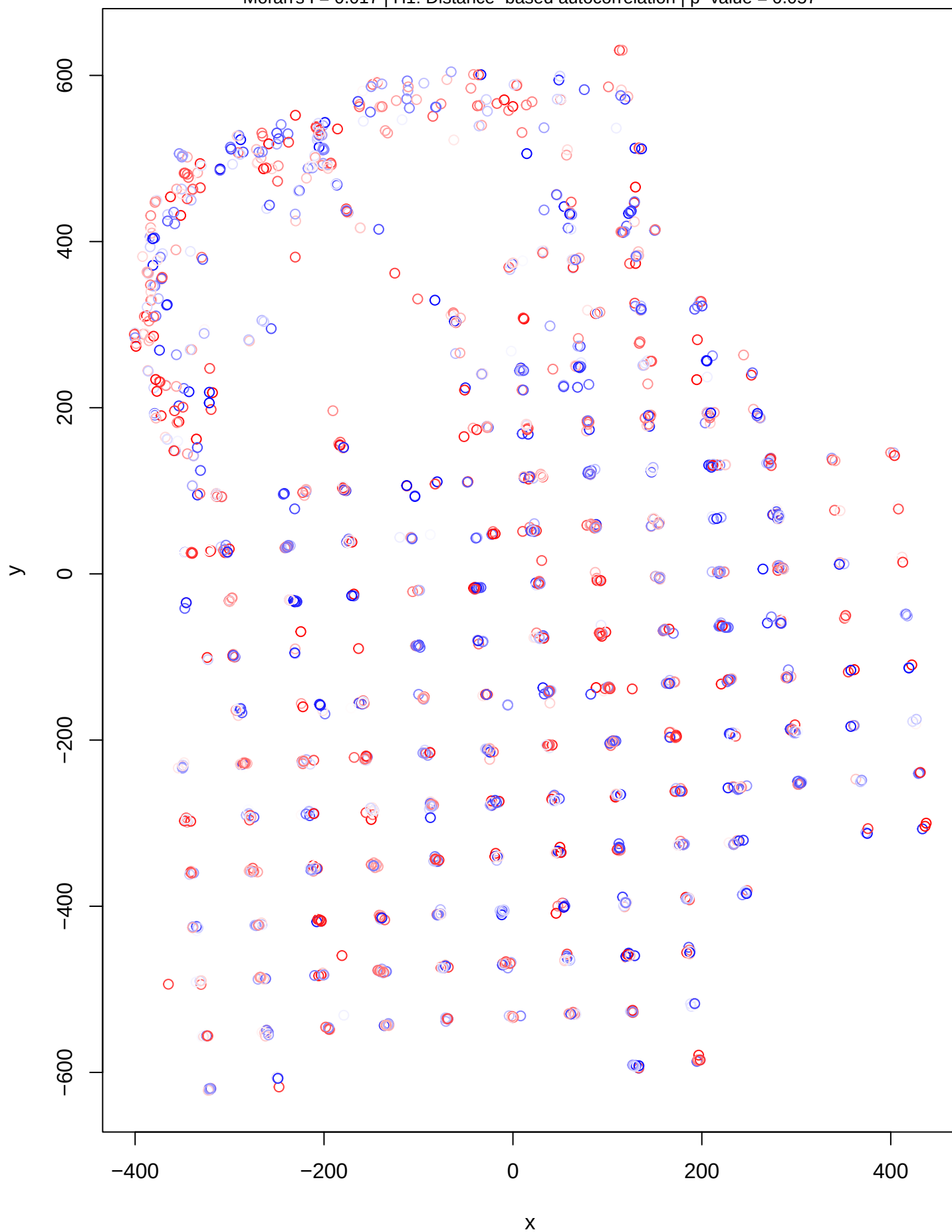
QQ plot residuals



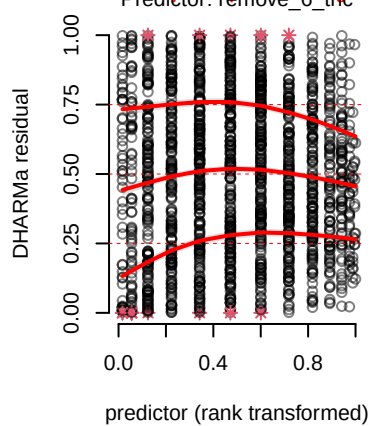
DHARMa residual vs. predicted
Quantile deviations detected (red curves)
Combined adjusted quantile test significant



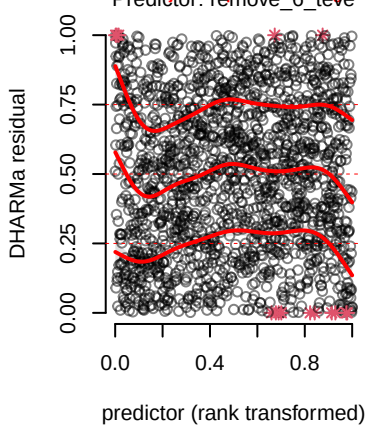
DHARMa Moran's I test for distance-based autocorrelation
Moran's I = 0.017 | H1: Distance-based autocorrelation | p-value = 0.057



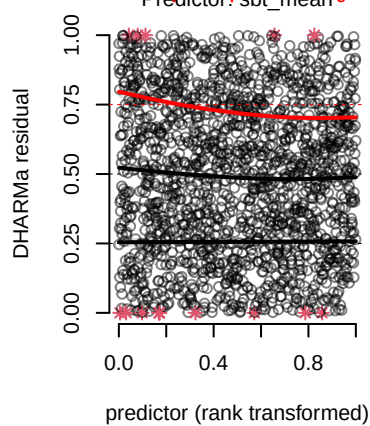
DHARMA residual Residual vs. predictor
Quantile deviations detected (red curves)
Combined adjusted quantile test significant



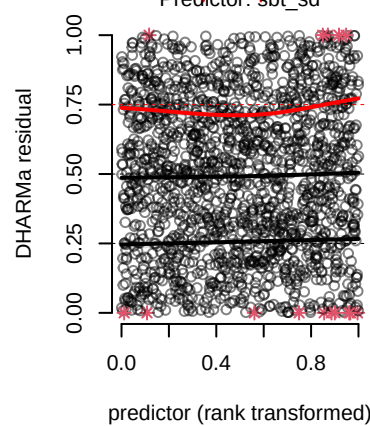
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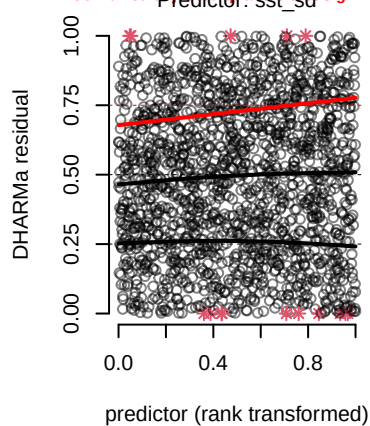
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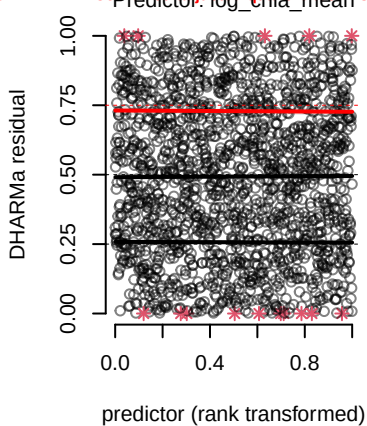
DHARMA residual Residual vs. predictor
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Combined adjusted quantile test n.s.



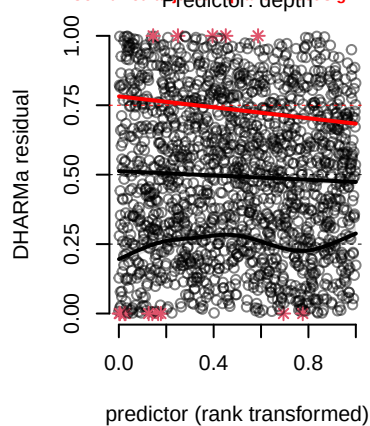
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